



Applications of Derivatives: Rectilinear Motion

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Identify position, velocity, and acceleration functions using derivative notation
- Compute total distance traveled and displacement of a particle over a time interval
- Determine when a particle changes direction by analyzing the velocity function

For each problem, use derivatives to analyze the motion of a particle and show all work in the space provided.

1. Given the position function below, find the velocity function $v(t)$.

$$s(t) = t^3 - 6t^2 + 9t$$

Answer: _____

2. Find the times when the particle is at rest for $s(t) = t^3 - 6t^2 + 9t$.

$$v(t) = 3t^2 - 12t + 9 = 0$$

Answer: _____

3. Find the acceleration function $a(t)$ for the position function $s(t) = t^3 - 6t^2 + 9t$.

$$a(t) = \frac{d^2s}{dt^2}$$

Answer: _____

4. Find the displacement of the particle on the interval from $t = 0$ to $t = 5$ seconds.

$$\text{Displacement} = s(5) - s(0)$$

Answer: _____

5. Find the total distance traveled by the particle during the first 5 seconds for $s(t) = t^3 - 6t^2 + 9t$.

$$D = |s(1) - s(0)| + |s(3) - s(1)| + |s(5) - s(3)|$$

Answer: _____

6. Find the speed of the particle at $t = 4$ seconds.

$$\text{Speed} = |v(4)|$$

Answer: _____

7. Determine whether the particle is moving forward or backward at $t = 2$ seconds.

$$v(2) = 3(2)^2 - 12(2) + 9$$

Answer: _____



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8. Find the acceleration at $t = 3$ seconds for $s(t) = t^3 - 6t^2 + 9t$.

$$a(3) = 6(3) - 12$$

Answer: _____

9. For $s(t) = t^3 - 6t^2 + 9t$, on which interval(s) is the particle moving backward during $0 \leq t \leq 5$?

$$v(t) < 0$$

Answer: _____

10. A particle has position $s(t) = t^3 - 6t^2 + 9t$. Find the total distance traveled during the first 3 seconds.

$$D = |s(1) - s(0)| + |s(3) - s(1)|$$

Answer: _____





Remind students that total distance requires breaking the interval at points where velocity equals zero, while displacement is simply $s(b) - s(a)$.

Solutions

1. Given the position function below, find the velocity function $v(t)$.

$$s(t) = t^3 - 6t^2 + 9t$$

- Take the derivative of $s(t)$ with respect to t .
- Apply the power rule to each term.
- The result is the velocity function $v(t) = 3t$ squared minus $12t$ plus 9 .

Answer: $v(t) = 3t^2 - 12t + 9$

2. Find the times when the particle is at rest for $s(t) = t^3 - 6t^2 + 9t$.

$$v(t) = 3t^2 - 12t + 9 = 0$$

- Set the velocity function equal to zero.
- Factor the quadratic as 3 times $(t$ minus 1) times $(t$ minus 3).
- Apply the zero product property to find t equals 1 and t equals 3.

Answer: $t = 1, t = 3$

3. Find the acceleration function $a(t)$ for the position function $s(t) = t^3 - 6t^2 + 9t$.

$$a(t) = \frac{d^2s}{dt^2}$$

- Differentiate the velocity function $v(t) = 3t$ squared minus $12t$ plus 9 .
- Apply the power rule to each term.
- The acceleration function is $a(t) = 6t$ minus 12 .

Answer: $a(t) = 6t - 12$

4. Find the displacement of the particle on the interval from $t = 0$ to $t = 5$ seconds.

$$\text{Displacement} = s(5) - s(0)$$

- Evaluate $s(5) = 125$ minus 150 plus 45 to get 20 .
- Evaluate $s(0) = 0$.
- Subtract to get displacement of 20 meters.

Answer: 20 meters

5. Find the total distance traveled by the particle during the first 5 seconds for $s(t) = t^3 - 6t^2 + 9t$.

$$D = |s(1) - s(0)| + |s(3) - s(1)| + |s(5) - s(3)|$$

- Identify the critical points at t equals 1 and t equals 3 where velocity equals zero.
- Compute $s(0) = 0$, $s(1) = 4$, $s(3) = 0$, $s(5) = 20$.
- Sum the absolute differences: 4 plus 4 plus 20 equals 28 meters.

Answer: 28 meters



6. Find the speed of the particle at $t = 4$ seconds.

$$\text{Speed} = |v(4)|$$

- Substitute t equals 4 into $v(t) = 3t$ squared minus $12t$ plus 9.
- Compute 48 minus 48 plus 9 to get 9.
- Take the absolute value to find the speed equals 9 meters per second.

Answer: 9 m/s

7. Determine whether the particle is moving forward or backward at $t = 2$ seconds.

$$v(2) = 3(2)^2 - 12(2) + 9$$

- Substitute t equals 2 into the velocity function.
- Compute 12 minus 24 plus 9 to get negative 3.
- Since velocity is negative, the particle is moving backward.

Answer: $v(2) = -3$, moving backward

8. Find the acceleration at $t = 3$ seconds for $s(t) = t^3 - 6t^2 + 9t$.

$$a(3) = 6(3) - 12$$

- Substitute t equals 3 into the acceleration function $a(t) = 6t$ minus 12.
- Compute 18 minus 12.
- The acceleration at t equals 3 is 6 meters per second squared.

Answer: 6 m/s²

9. For $s(t) = t^3 - 6t^2 + 9t$, on which interval(s) is the particle moving backward during $0 \leq t \leq 5$?

$$v(t) < 0$$

- Use the factored velocity $v(t) = 3(t \text{ minus } 1)(t \text{ minus } 3)$.
- Test a value such as t equals 2 to see that v is negative.
- Conclude the particle moves backward on the open interval from 1 to 3.

Answer: $1 < t < 3$

10. A particle has position $s(t) = t^3 - 6t^2 + 9t$. Find the total distance traveled during the first 3 seconds.

$$D = |s(1) - s(0)| + |s(3) - s(1)|$$

- Identify the critical point at t equals 1 within the interval from 0 to 3.
- Compute $s(0) = 0$, $s(1) = 4$, $s(3) = 0$.
- Sum absolute differences: 4 plus 4 equals 8 meters.

Answer: 8 meters

